Taylor series

For a function $f(x)$ of a single variable, if it has a Taylor series, expanded around the base point 0, of the form:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

where $f^{(n)}$ is the nth - derivative of f. i.e. $\frac{d^n f}{dx^n}$, which is valid in some interval containing 0, potentially all of $\mathbb{R}$.

Partial sums of this provide approximations for f, i.e.

$$f(x) \approx \sum_{n=0}^{k} \frac{f^{(n)}(0)}{n!} x^n$$

which get better and better as $k \to \infty$.

What do Taylor series look like for functions of multiple variables.

Consider a function $f(x, y)$ and a base point (a, b). We'll write $h = \Delta x$, $k = \Delta y$, for small variations in x, y. Then the Taylor series for $f(x, y)$

about (a,b) will be.

$$f(a+h,b+k) = \sum_{n=0}^{\infty} \frac{1}{n!} (D^n f)(a,b)$$

where D is the differential operator

$$D = h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}$$

, and the powers D^n are the compositions of D with itself.

ie. $D^n f = D(D(D(\ldots(Df))\ldots))$

(n copies of D)

The first few instances of $D^n f$.

$$D^0 f = f$$

$$Df = h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}$$

$$D^2 f = D(Df) = \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)\left(h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}\right)$$

$$= h^2\frac{\partial^2 f}{\partial x^2} + k^2\frac{\partial^2 f}{\partial y^2} + 2hk\frac{\partial^2 f}{\partial x \partial y}$$

(remember Clairaut's theorem $\partial^2 f/\partial x\partial y = \partial^2 f/\partial y\partial x$)

and

$$D^3 f = h^3 \frac{\partial^3 f}{\partial x^3} + 3h^2 k \frac{\partial^3 f}{\partial x^2 \partial y} + 3hk^2 \frac{\partial^3 f}{\partial x \partial y^2} + k^3 \frac{\partial^3 f}{\partial y^3}$$

and in general

$$D^n f = \sum_{j=0}^{n} \binom{n}{j} h^j k^{n-j} \frac{\partial^n f}{\partial x^j \partial y^{n-j}}$$

where $\binom{n}{j} = \frac{n!}{j!(n-j)!}$

Example Consider (a, b)

$f(x, y) = \sin(x + 3y) + \cos(3x + y)$ //

Find the Taylor expansion around $(\pi/2, 0)$

up to order $n = 2$.

We expect to find

$f(\pi/2 + h, k) \approx$ polynomial in h, k to order 2.

$$(D^0 f)(\pi/2, 0) = f(\pi/2, 0)$$

$$= 1$$

$$(D^1 f)(\pi/2, 0) = \left(h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y} \right)_{(\pi/2,0)}$$

$$= \Big(h(\cos(x+3y) - \sin(3x+y)3) + k(3\cos(x+3y) - \sin(3x+y)) \Big)_{(\pi/2,0)}$$

$$= 3h + k$$

For $D^2 f$ we'll need the second order derivatives

$$\frac{\partial^2 f}{\partial x^2} = -\sin(x+3y) - 9\cos(3x+y) = -1, \text{ at } (\pi/2, 0)$$

$$\frac{\partial^2 f}{\partial y^2} = -9\sin(x+3y) - \cos(3x+y) = -9, \text{ at } (\pi/2, 0)$$

$$\frac{\partial^2 f}{\partial x \partial y} = -3\sin(x+3y) - 3\cos(3x+y) = -3, \text{ at } (\pi/2, 0)$$

$$\text{So } (D^2 f)(\pi/2, 0) = -h^2 - 9k^2 - 6hk$$

Putting all this together, claims that f is approximated by.

$$f(\pi/2 + h, k) \approx 1 + 3h + k - \frac{h^2}{2} - \frac{9}{2}k^2 - 3hk$$

Let's visualize this on the computer.

This will generalize to to any number

n variables easily.

say $f(x_1, \ldots, x_m)$

$$f(a_1+h_1, a_2+h_2, a_3+h_3, \ldots, a_m+h_m)$$

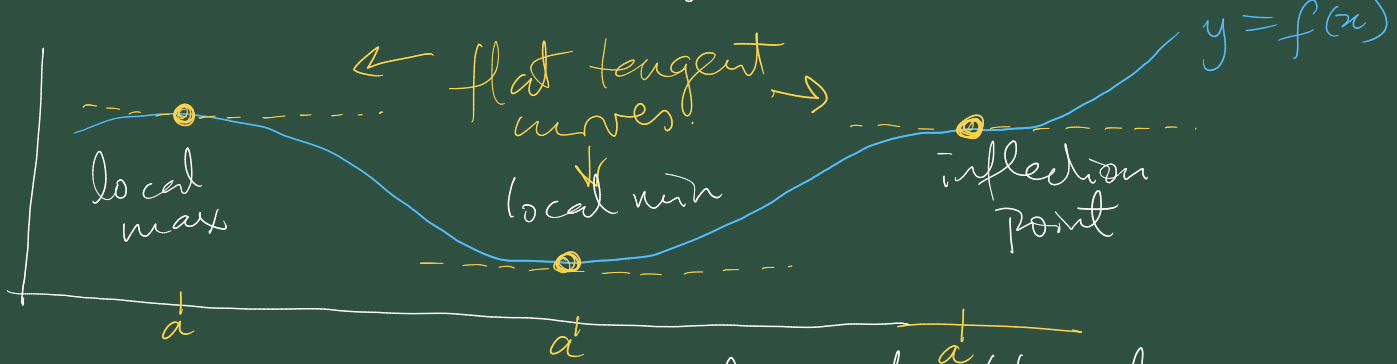
$$= \sum_{n=0}^{\infty} \frac{(D^n f)}{n!}(a_1, \ldots, a_m)$$

where $D = \sum_{i=1}^{m} h_i \frac{\partial}{\partial x_i}$

(Continue at 12:05.)

Optimisation in 2 variables

Finding and classifying critical points of a 1-variable function $y = f(x)$



The three types of critical point (local max, local min, inflection point) are found by the ~~solutions~~ solutions to

$$f'(x) = 0$$

and classified using the second derivative

$f''(x)$

$f''(a) > 0$, a is a local min.

$f''(a) < 0$, a is a local max.

$f''(a) = 0$, a is an inflection point.

In the two-variable case we are considering the surface defined by

$$z = f(x, y)$$

Def A point (a, b) is a critical point for f if and only if.

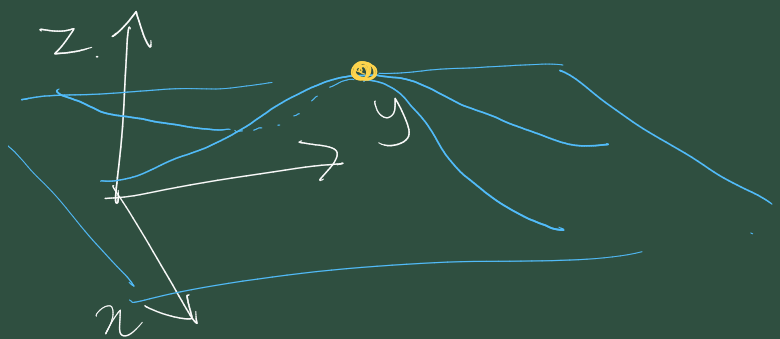
AND.

$$\frac{\partial f}{\partial x}(a, b) = 0 \quad \& \quad \frac{\partial f}{\partial y}(a, b) = 0$$

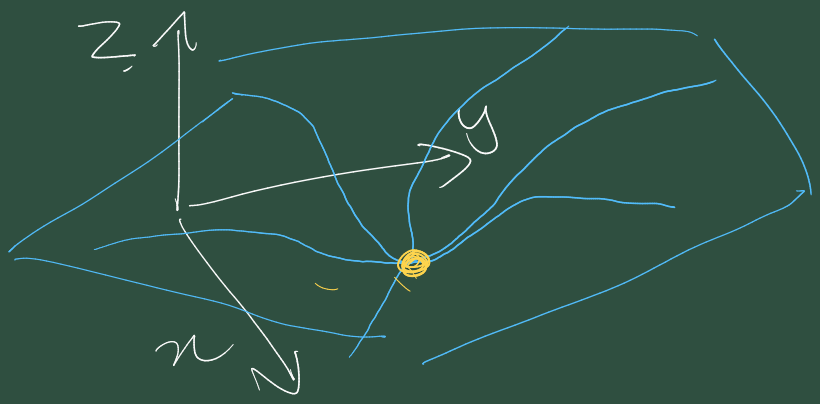
Geometrically this corresponds to a flat tangent plane to the surface at a, b.

Again they come in three types

- local max ("top of a hill")

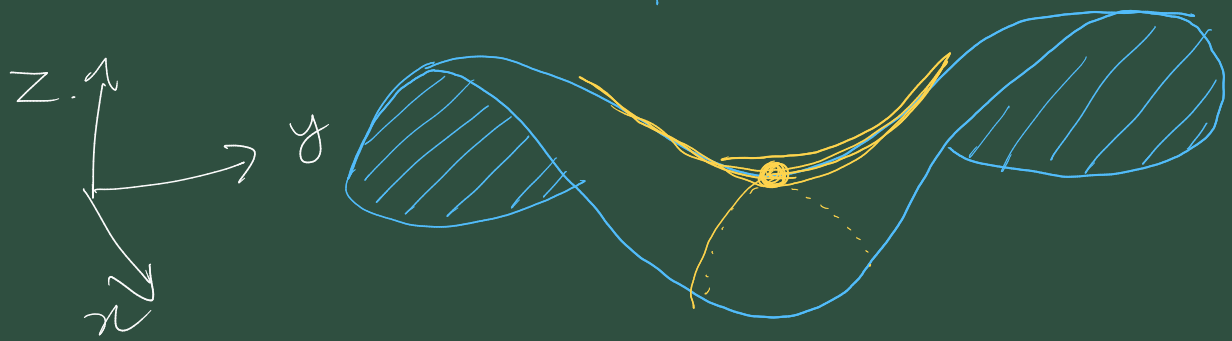


- local min ("bottom of a bowl").



- saddle point.

shading the bottom side of the surface.



These are classified using 2nd order derivative information by the Hessian determinant.

$$D = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix}$$

$$= \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

If $D(a,b) < 0$ then (a,b) is a saddle,

and if $D(a,b) > 0$

then $\begin{cases} \frac{\partial^2 f}{\partial x^2}(a,b) > 0 \Rightarrow (a,b) \text{ is a local minimum.} \\ \frac{\partial^2 f}{\partial x^2}(a,b) < 0 \Rightarrow (a,b) \text{ is a local maximum.} \end{cases}$

This can be justified by an analysis of the 2nd order Taylor polynomial for f.

Example find and classify critical points of

$$f(x,y) = x^3 + 3xy^2 - 15x - 12y$$

$$\frac{\partial f}{\partial x} = 3x^2 + 3y^2 - 15$$

$$\frac{\partial f}{\partial y} = 6xy - 12$$

To find the simultaneous solution, consider the simpler equation first.

$$\frac{\partial f}{\partial y} = 0 \iff 6xy - 12 = 0.$$

$$\iff xy = 2.$$

$$\iff y = \frac{2}{x}, \text{ not we know } x, y \neq 0 \text{ at a critical points.}$$

Then impose the solution condition onto the other equation.

a simultaneous solution will be a solution of

$y = \frac{2}{x}$ AND $\frac{\partial f}{\partial x} = 0$

$\Leftrightarrow \quad 3x^2 + 3\left(\frac{2}{x}\right)^2 - 15 = 0$

$\Leftrightarrow \quad 3x^2 + \frac{12}{x^2} - 15 = 0.$

$\Leftrightarrow \quad 3x^4 - 15x^2 + 12 = 0$

$\Leftrightarrow \quad x^4 - 5x^2 + 4 = 0.$

$\Leftrightarrow \quad (x^2 - 4)(x^2 - 1) = 0.$

$\Leftrightarrow \quad x = -2, 2, -1, 1$

Giving the four critical points

$(-2, -1), (2, 1), (-1, -2), (1, 2)$

Need the Hessian determinant:

$$D = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

$$= 6x\,6x - (6y)^2$$

$$= 36x^2 - 36y^2$$

$$= 36(x^2 - y^2)$$

So

$D(-2, -1) = D(2, 1) = 36 \cdot 3 = 108 > 0$

so local mins/maxs

$$\frac{\partial^2 f}{\partial x^2} = 6x \begin{cases} < 0 & \text{at } (-2,-1) \\ > 0 & \text{at } (2,1) \end{cases}$$

So $(-2,-1)$ is a local maximum and

$(2,1)$ is a local minimum

AND $D(-1,-2) = D(1,2) = -3 \cdot 36 = -108 < 0$

so $(-1,-2)$, $(1,2)$ are both saddle points.

Let's repeat/verify this on Matlab.